Algorithmic bias in a social policy data tool for environmental justice

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Abstract

Neighborhood-level screening algorithms are increasingly being deployed to guide policy decisions regarding environmental justice. We evaluate one such algorithm, used to allocate hundreds of millions of dollars in public funding annually, and compare it against plausible alternate specifications to identify flaws in model behavior. The model underweights neighborhoods with extreme values of pollution or health indicators, and also exhibits algorithmic bias against foreign-born populations, leading to large funding differences compared to a model with small corrections applied (AUC = 0.87, corresponding to an annual funding difference of $81M). We also find that in the case of usage by a malicious actor, small manipulations to the model can divert funding based on race and political affiliation. We provide a number of technical recommendations, and urge that tools such as this should be designed by and in collaboration with community leaders, domain experts, and regulatory agencies in order to avert misallocation of funding and exacerbation of social inequities.

Background

Algorithms based on high-resolution data are being increasingly used for a wide range of high-impact use-cases, including policymaking (cite). The use of area based measures for allocating funds has advantages over individual measures in that they are less likely to be subject to individual manipulation, generally have lower administrative costs, and are focused on community programs rather than households. Many such algorithms, however, have fallen under scrutiny, as audits of them have yielded evidence of algorithmic bias against structurally marginalized populations (cite). Here we investigate a data tool from the state of California that designates neighborhoods eligible for social services funding, motivated by addressing concerns of environmental justice.

History

In 2012, California Senate Bill 535 was signed into law, allocating 25% of California's proceeds from its cap-and-trade program to social welfare funds for environmentally disadvantaged communities. CalEnviroScreen, an open-source data tool, was created by CalEPA to identify and designate disadvantaged communities.[1] The tool directs an estimated $800 million of funding annually from California's cap-and-trade program alone; it also directly influences funding from a variety of public and private sources, and is reported to have directed an estimated $12.7 billion in funding.[2] The funding targets of the tool are varied, including funding

[1] https://oehha.ca.gov/calenviroscreen
[2] https://www.sfchronicle.com/bayarea/article/California-has-a-tool-to-fund-its-most-vulnerable-16643115.php

programs for affordable housing, land use strategies, agricultural subsidies, wildfire risk reduction, public transit, and renewable energy. Similar data tools are in use or development at the federal and state levels across the United States of America.

Overview of data tool

The tool takes Census tract-level data on pollution burden and population characteristics and reduces it into a single score; the Census tracts in the top 25% of scores are designated "disadvantaged communities." The pollution burden category is split into two subcategories: exposures and environmental effects; the population characteristics category is also split into two subcategories: socioeconomic factors and sensitive populations. All categories are weighted equally except for environmental effects, which is half-weighted. Variables in each subcategory are converted to ranked percentiles then averaged together into a subcategory score. The subcategory scores are then averaged into category scores (pollution burden and population characteristics), which are then multiplied together to create a CalEnviroScreen (CES) score. (Figure 1a)

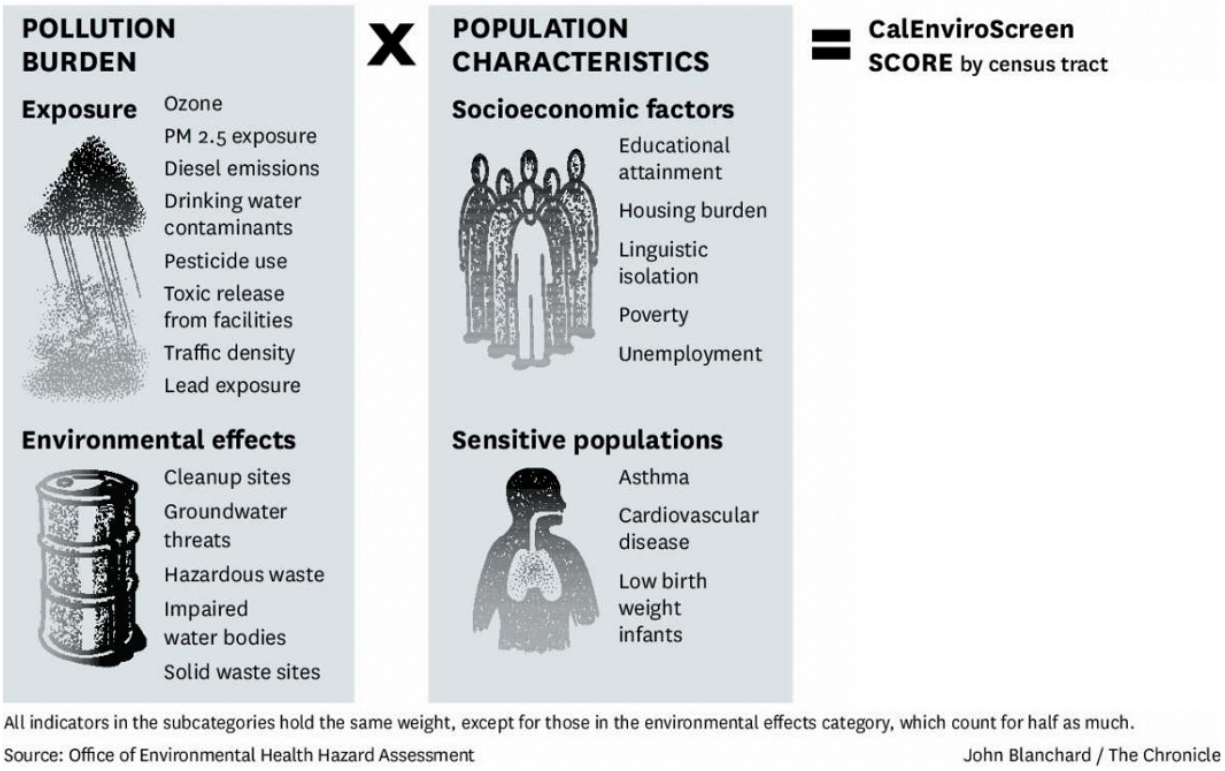


Figure 1a: An overview of CalEnviroScreen's methodology. (to be replaced)

Composite indicators such as the scoring system provided by CalEnviroScreen do not reflect objective truths, but are the result of subjective design choices regarding variable choice, normalization, and aggregation (cite Saisana, Kuc-Czarneka). Policy decisions such as funding allocation resulting from composite indicators is thus intrinsically subject to human and

algorithmic bias at the development and deployment stage. Accordingly, we perform an audit of this environmental justice data tool through the lens of public health and algorithmic fairness, illustrating potential pitfalls in its usage. We examine how much output from the tool can change based on subjective choices in five different categories: (1) its conceptual definition, (2) its measurement choices, (3) its mathematical definition, (4) its decision threshold, and (5) its potential misuse by malicious actors.

Conceptual considerations for inclusion of domains

Considering that CalEnviroScreen has been used to fund a diverse range of applications, not all domains may be applicable for each funding target (Figure 1a, 1b). For example, pollution may not be the most valuable domain for determining where to allocate affordable housing or agricultural subsidies. We calculated the model difference from removing a given domain, and found that 7.4-16.9% of designated census tracts change based on removing a domain, equivalent to $59-135M of funding. This sensitivity of the model to removing domains suggests that careful consideration may be needed as to whether a single composite indicator is appropriate for allocating funding across different sectors.

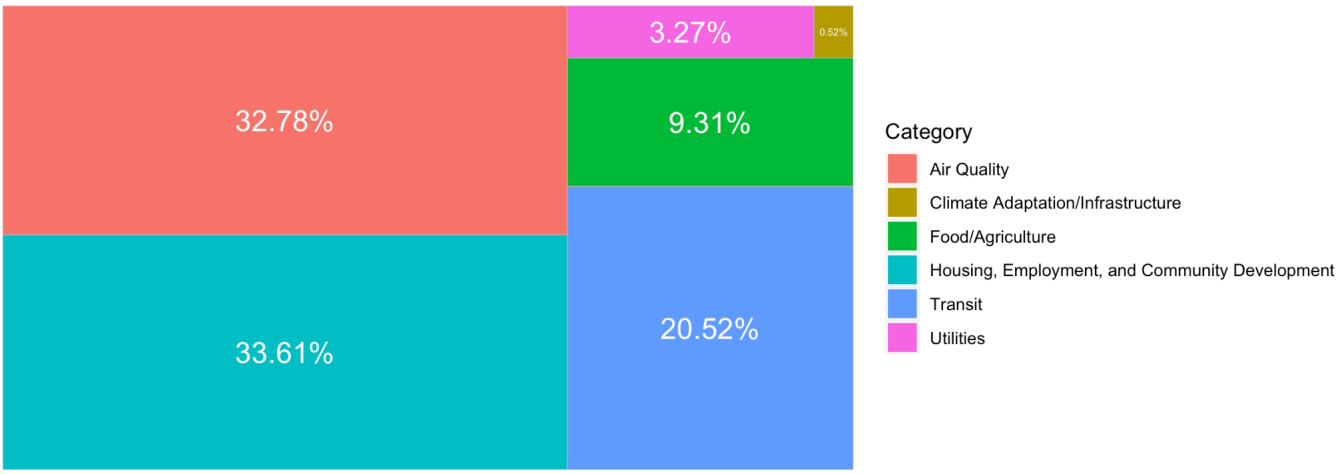


Figure 1b: Funding allocation to tracts designated as disadvantaged communities by the CalEnviroScreen model. Colors indicate different categories of funded projects. Cumulative funding amount is $2.6bn from 2015 to 2021.

Model specification

Subjective model specifications such as variable choice, normalization, aggregation, and weighting lead to substantial changes in the CalEnviroScreen model output. Our proposed set of model specifications differs from the original model by 9%, equivalent to an annual funding difference of $67M. Through sensitivity analysis, we observe that 23% of tracts originally

designated as disadvantaged can lose their designation from plausible alternative models (Figure 2a).

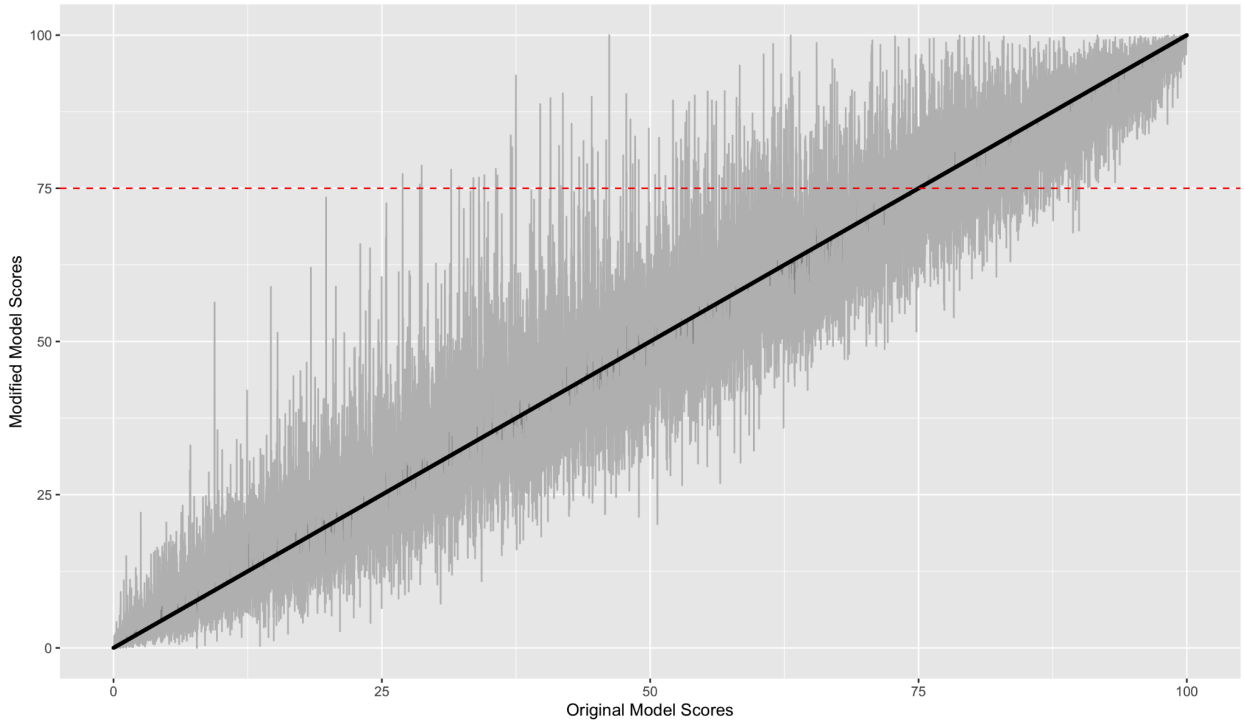


Figure 2a: Model sensitivity to input parameters. Black diagonal indicates CES scores from original CES model, gray bars indicate maximum and minimum values from alternative plausible model specifications, and dashed red line indicates the cutoff score for funding designation.

Variable choice and measurement errors

Subjective decisions in measuring, including, or excluding variables from the model can have outsized impacts on the final model output. For example, the tool measures cardiovascular and respiratory health in terms of emergency department visits for heart attacks and asthma attacks. These two variables are poorly correlated with a plausible alternative – model-based estimates derived from survey data on cardiovascular (R^2=0.05) and respiratory health (R^2=0.26) – indicating large differences between the two measurements.

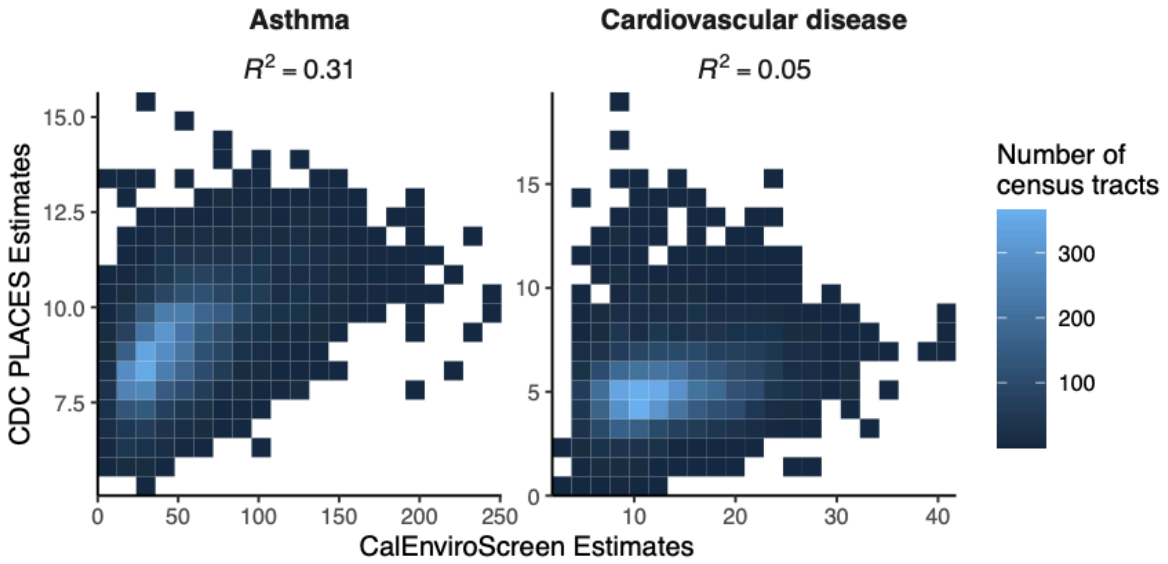


Figure 2b: Low correlations between Census-tract level ER visits (asthma and heart attacks) and survey data (history of asthma and coronary heart disease).

Furthermore, the choice of asthma over other respiratory diseases, such as chronic obstructive pulmonary disease (COPD), biases against communities with large foreign-born populations. Asthma, which is the only respiratory disease included in the model, is less prevalent among recent immigrant populations compared to people born in the United States (cite). We find that as census tracts increase in percent foreign born, census tracts are less likely to be classified as disadvantaged under the asthma model compared to the COPD model, diverting less funding to communities with large immigrant populations.

Lastly, the current model measures overall population health in terms of low birth weight, cardiovascular health, and respiratory health – an incomplete measure of populations vulnerable to environmental exposures. Overall, we find that models incorporating a wider range of health metrics known to be indicators of vulnerability to environmental exposures, such as cancer, kidney disease, COPD, or using survey data in addition to emergency department data, results in a 2.2-4.6% change of designated census tracts (equivalent to $18-37M of funding).

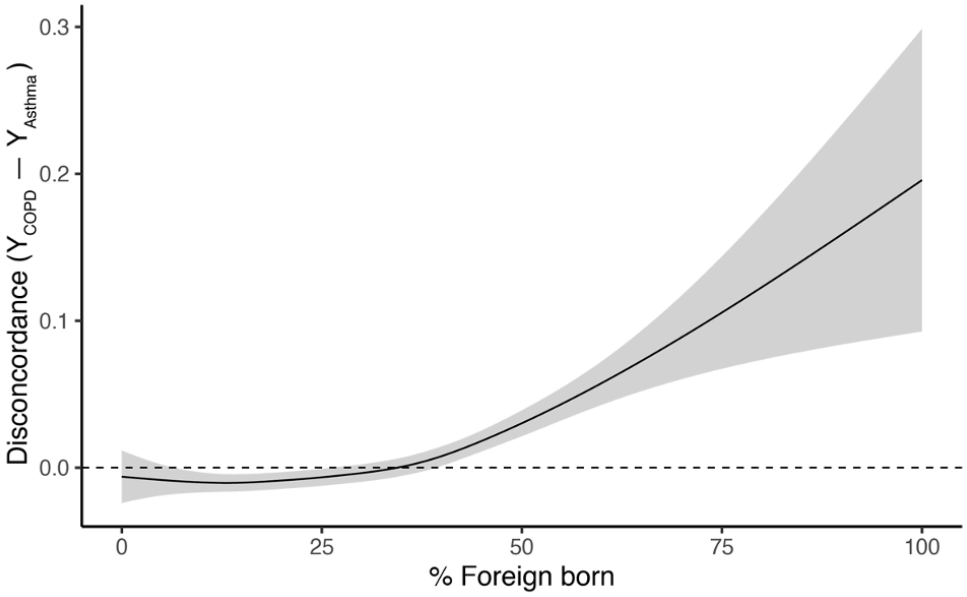


Figure 2e: Discordance between the original model using asthma as an indicator for respiratory health and a new model using COPD instead. The models are essentially the same except for tracts that have more than a 30% foreign-born population, suggesting model bias against tracts with high immigrant populations.

Normalization and aggregation

We found that a small change in pre-processing variables – switching from percentile ranking to z-score standardization – leads to a 5.3% change in designated census tracts (equivalent to $43M of funding). The CalEnviroScreen metric uses percentile ranking to pre-process variables, which converts the original distributions of the raw data to uniform distributions: extreme values become underweighted and values near the center of the distribution become overweighted.

For example, the difference between the 60th and 80th percentile values of measurements across tracts is relatively low (a median difference of .6 standard deviations), whereas the difference between the 80th and 99th percentiles is high (median difference of 2.6 standard deviations). Pre-processing through percentile ranking would erroneously equate these two differences.

Similarly, we find that changing the method of aggregating the two subtotal values – from multiplicative to additive – leads to a 1.4% change in designated census tracts (equivalent to $11M of funding). Multiplication is less compensatory than additive aggregation: under multiplicative aggregation, low scores in one category cannot be fully compensated for by high scores in another category, thus encouraging above-average values in both categories (cite Saisana 2005). For example, a tract with pollution burden and population characteristic scores of (9.2, 4.3) does not make the cutoff, yet a tract with scores of (6.5, 6.1) does. Considering the small difference between scores closer to the center of the distribution, additive aggregation is likely more appropriate for the data.

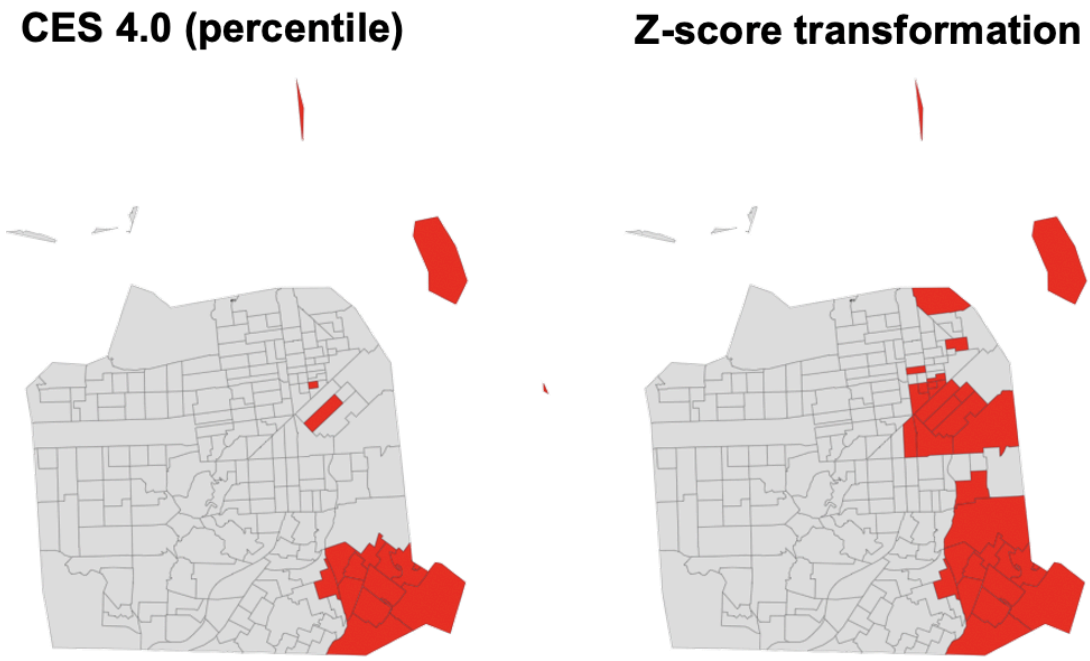
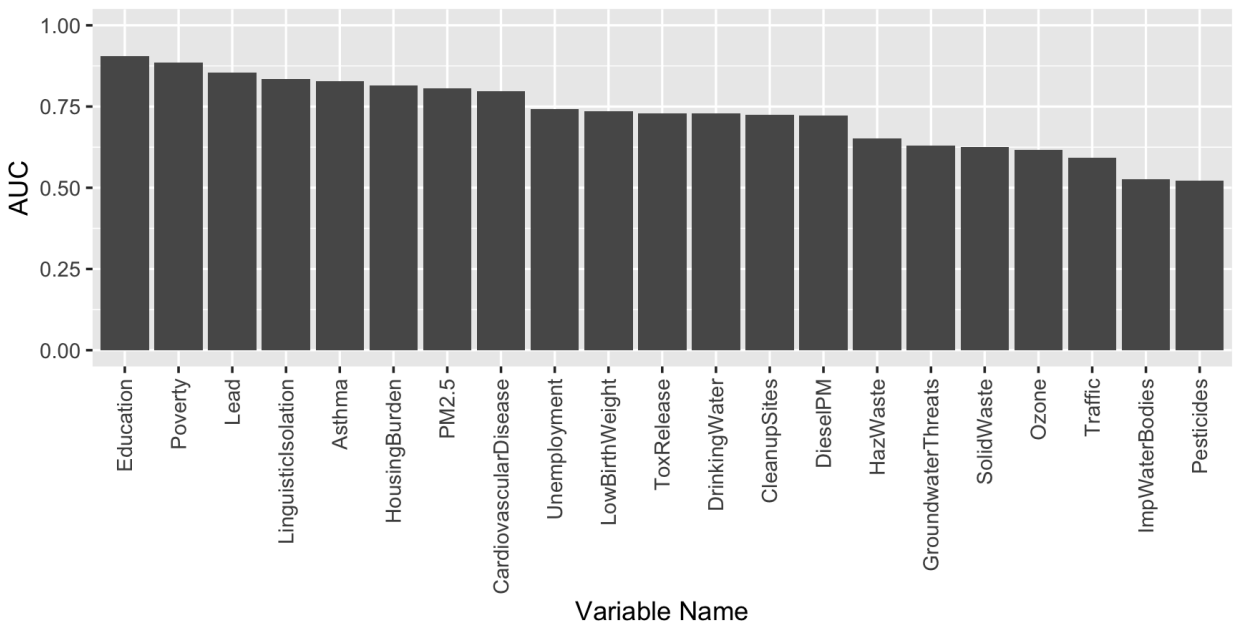


Figure 2d: Census tracts in San Francisco, CA. Red shading indicates DAC designation under different normalization regimes.

Weighting

Many variables in CalEnviroScreen's model are strongly correlated with each other (Figure 2e), yet each are weighted almost entirely equally (with the exception of environmental effects variables) – consequently, each variable's ability to predict DAC designation is almost entirely explained by its level of correlation to the other variables (R^2 = 0.96). Highly correlated variables like education levels can single-handedly predict DAC designation to an accurate degree (AUC = 0.91). Conversely, environmental variables like pesticide exposure or impaired water bodies have negligible impact on the final model. The existing weighting scheme is a subjective model choice with the impact of implicitly overweighting the importance of socioeconomic indicators at the expense of environmental exposures and effects. An alternative weighting scheme using principal components analysis results in an X% change in census tracts, equivalent to $Y in funding.

Careful consideration should be given towards how to weight each variable. If it is intended for education or poverty to be near-perfect proxies for the CES score, it needs to be justified with pre-specified theoretical foundations. Post-hoc justifications for weighting schemes could otherwise be used to manipulate the algorithm for various means.


AUC
1.00
0.75
0.50
0.25
0.00
Education
Poverty
Lead
LinguisticIsolation
Asthma
HousingBurden
PM2.5
CardiovascularDisease
Unemployment
LowBirthWeight
ToxRelease
DrinkingWater
CleanupSites
DieselPM
HazWaste
GroundwaterThreats
SolidWaste
Ozone
Traffic
ImpWaterBodies
Pesticides
Variable Name

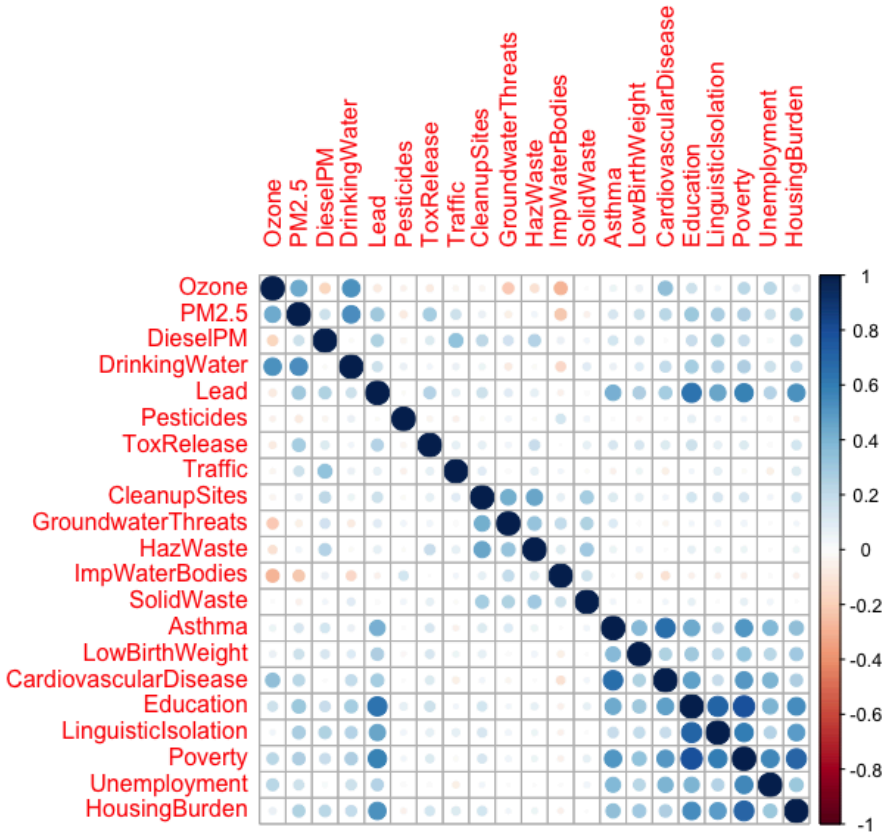

Ozone
PM2.5
DieselPM
DrinkingWater
Lead
Pesticides
ToxRelease
Traffic
CleanupSites
GroundwaterThreats
HazWaste
ImpWaterBodies
SolidWaste
Asthma
LowBirthWeight
CardiovascularDisease
Education
LinguisticIsolation
Poverty
Unemployment
HousingBurden
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

Figure 2e: Variable AUC and cross-correlation matrix.

Post-deployment considerations

Composite indicators are typically used for exploratory analyses rather than tools for policymaking or immediate impact (cite). CalEnviroScreen's identity as a funding designation tool creates a scenario where parties in control of it have outsize influence on where funding should go.

Identifying adversarial misuse

Alarmingly, we find that due to the model's sensitivity to subjective design choices, should a malicious actor gain control of the algorithm, small manipulations can be made to divert funding to neighborhoods based on race or political affiliation. Using a constrained optimization approach, we induced minor changes to the weights of variables, resulting in a 36% increase in neighborhoods receiving funding based on political affiliation, as well as a 25% increase based on race. Therefore in the absence of regulatory guidelines, plausible subjective changes to the model can be exploited for political gain.

Decision threshold

The model's hard funding cutoff at the 75th percentile means neighborhoods just above the threshold are eligible for funding, and those just below are ineligible. Given the underlying model's sensitivity to subjective parameter choices, census tracts just below the threshold may be seen as similarly disadvantaged as those just above the threshold – a soft threshold funding model would provide funding more equitably. Funding agencies charged with interpreting output from composite indicators should take care to consider observations close to the cutoff point, and how they might change through sensitivity analyses. (Figure 2f)

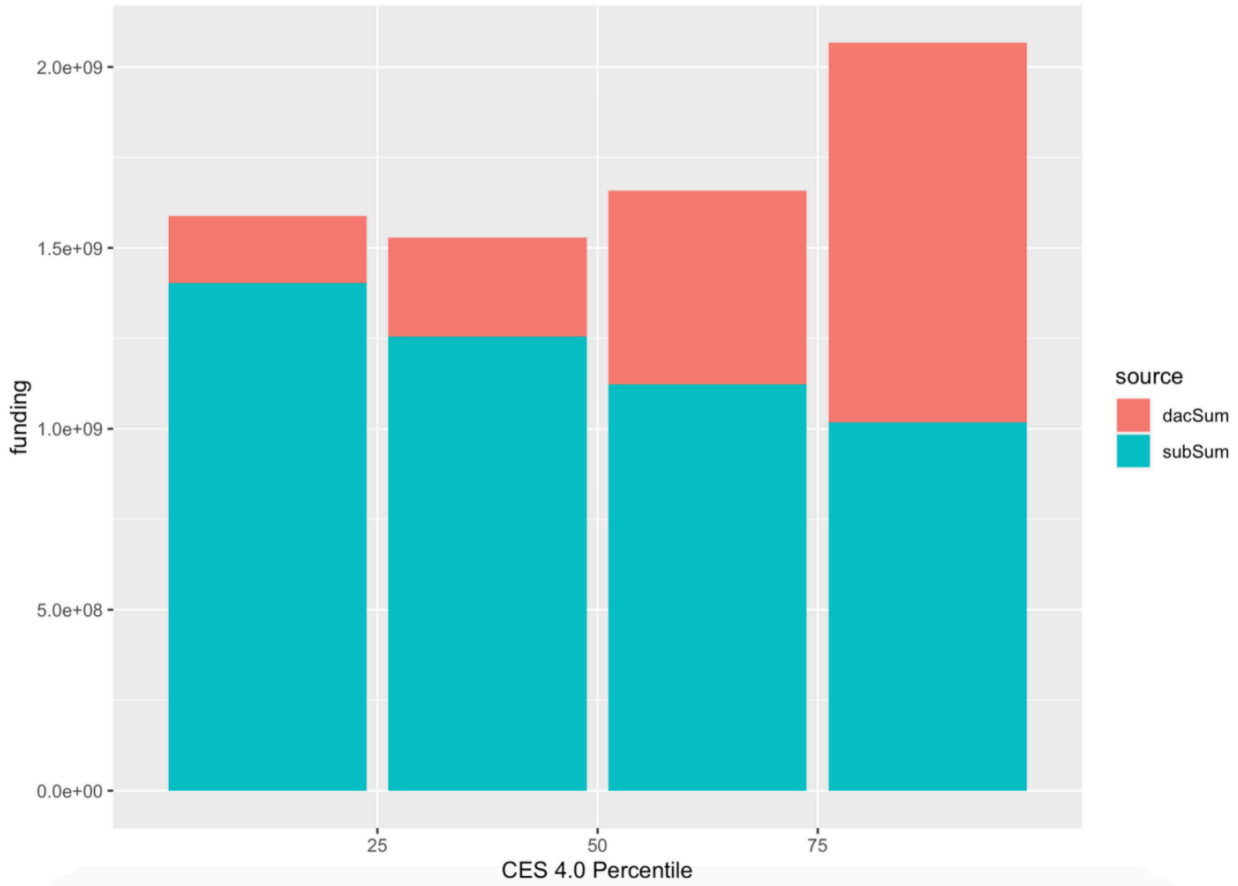


Figure 2f: Total cumulative funding by California Climate Investments implemented in census tracts by CES percentile. Red bars indicate funding specifically for disadvantaged communities; blue bars indicate all other funding.

Discussion

Our results show that CalEnviroScreen's algorithm, used for funding allocation, is potentially flawed along five different axes: conceptual definitions, variable choice, mathematical formulation, algorithmic fairness, and adversarial misuse. The limitations of our work are largely the same limitations besetting the data tool itself: key missing indicators, such as indoor air pollution or drinking water quality, a lack of ground-truth designation for a disadvantaged community, and varying levels of data quality. Accordingly, we do not comprehensively identify every potential flaw of the tool in our analysis, but instead highlight several examples to illustrate how sensitive the model is to changes in its definition.

Some of the issues we raise have technical solutions, for which we offer the following recommendations: (1) use z-score standardization instead of percentile-ranking to pre-process the variables, thereby preserving the distributions of the raw data; (2) use averaging instead of multiplication to calculate the total score in order to avoid amplifying existing model biases; (3)

expand the Sensitive Populations category to include more medical variables and data sources, thereby being more broadly inclusive and mitigating potential bias associated with any particular variable; (4) use a soft threshold instead of a hard threshold to designate neighborhoods eligible for funding, thereby more fairly allocating funding to neighborhoods close to the decision threshold; and (5) use sensitivity analyses to assess robustness to model specifications (cite Saisana 2005).

On a fundamental level, however, we argue the screening tool should narrow its scope. First, a screening tool based on available data inherently provides an incomplete picture of environmental justice concerns for any given community, especially as poorer communities tend to have fewer and lower-quality data (cite). Considering that any number of arbitrary model decisions could substantially sway the results of the algorithm, funding allocated based on the tool is thus determined in a volatile manner, and disproportionately so for poorer communities. Given these severe limitations, screening is useful as a part of an initial exploratory phase, but should not be used as a final arbiter of funding allocations.

To that end, we recommend the tool include a mechanism for accountability in the form of community engagement and regulatory oversight. First, consistent input from community members, domain experts, and stakeholders should guide the designation of disadvantaged communities. As it is now, members of the public may provide comments, but are not explicitly involved in decision-making regarding the development of the tool. The agency governing the tool has unilateral authority to modify the algorithm; there is therefore an alarming potential for an internal actor to exploit the algorithm for political or financial gain. Safeguards, such as mandating transparency in model development or an independent review committee, need to be in place to prevent misuse.

Methods (under construction)

Prior to data analysis, we identified potential flaws in the data tool and conceived plausible alternative models. We reproduced the original CalEnviroScreen model based on its documentation, and validated our reproduction. As a general approach, we built alternative models and evaluated how they differed from the original model. Details of each step of this approach are below.

Data

We used census tract-level data obtained from CalEnviroScreen, consisting of 8035 observations with 21 variables measuring different aspects of environmental exposures and population characteristics. For our additional variable analyses, we used the PLACES dataset (cite) to include tract-level variables on estimated prevalence for asthma, cancer, chronic kidney disease, and coronary heart disease.

Pre-processing errors

We recreated the CalEnviroScreen model with a modified normalization procedure: using Z-score standardization instead of percentile ranking as a way of converting each variable onto the same scale before converting into an index. We tested each variable

Additionally, we propose averaging the pollution burden and population characteristics together to create the CES score instead of multiplying them. Multiplying the two category scores inflates high values and deflates low values from the category scores, exacerbating existing model error.

Measurement error

Currently, the data tool uses emergency room visits for asthma and myocardial infarction as proxies for respiratory and cardiovascular health, respectively. Using ER visits as a metric is potentially biased: although on average, lower-income populations use the ER disproportionately more, some subgroups, such as immigrant populations, use the ER disproportionately less.[3] We propose using a mix of ER visits and Census tract-level survey data for a more robust estimate of respiratory and cardiovascular health.

Model design

The inclusion and exclusion of variables to use in the model is inherently a subjective choice and can result in inadvertent model bias. For example, we identify the use of asthma as a representative of respiratory health as a potentially biased indicator. Because asthma is less prevalent in developing countries, it is disproportionately less prevalent amongst immigrant populations in the US[4]; consequently, the model is potentially biased against neighborhoods with large immigrant populations. As an alternative, we propose using a mix of asthma and chronic obstructive pulmonary disease (COPD) as an indicator for respiratory health.

Additionally, respiratory and cardiovascular health are not the only areas of health affected by environmental exposures. We propose using more indicators, such as renal function and neurological disorders.

Evaluation

We evaluate model agreement between the original model and alternative models using a variety of metrics ranging from area under the ROC curve, precision-recall, Cohen's Kappa, etc. We also evaluate heterogeneity in model agreement using discordance as a metric: if discordance between two models increases as a protected attribute (e.g., race, age, etc) also increases, then the difference in models presents as bias against the protected attribute.

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[3] https://pubmed.ncbi.nlm.nih.gov/23447058/
[4] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4011106/

Figures

To-do

- Do figures
    - Figure 1: intro figure
        - 1a: how CES algorithm works
            - break down algorithm design
            - show which categories go where
        - 1b: where funding goes
            - how is funding divvied up, etc.
        - 1c (?): introduce our five categories of results?
        - maybe:
            - 1d (?): our proposed solution?
                - Infographic showing conceptual framework for how to build a data-informed tool for policy making
            - 1e (?): example of how tool works poorly?
                - e.g., SF vs LA map
    - Figure 2: results figure
        - 5 parts that reference 1c
        - Maybe put 1d here (proposed solution)
        - composite figure that shows all of our proposed changes
            - AUC differences?
                - ROC plot with multiple curves, each one representing different sets of proposed changes?
                - consider doing 1-ROC? lower AUC = better is a weird concept.
            -